# JIANT FM · SOUND · FX · MOD

engines, effects, modulation, sequencer and songs

firmware v0.6.32 · M-VAVE FM-1 · juanjiant-bit.github.io/JIANT

## ENGINES · EDIT 1 / EDIT 2

### EIGHT ENGINES, ALL SYNTHESIS

| ENGINE | EDIT 1 | | | | EDIT 2 | | | | WHAT TO KNOW |
|---|---|---|---|---|---|---|---|---|---|
| ANALOG | WAVE | DETUNE | MIX | NOISE | CUT | RES | DRIVE | INT | SAW SQR TRI SIN PWM **SYNC RING SAW3**; INT: osc 2 ±24 st. Its filter is per voice. |
| FM6 | ALG | FB | MOD | RATIO | MEG | VMOD | DETUNE | SLOT | Dexed's 6 operators, 32 algorithms drawn, .syx patches (SLOT), MOD / RATIO move the modulators. **OPERATOR · OP ENV · OPERATOR 2**: each operator on the device (ratio / fixed, fine, level, its envelope, mode, detune, velocity). **SLOT B1–B32**: a 32-voice .syx sent from the web editor. |
| PHASE | WAVE | WAVE2 | DCW | ENV | DETUNE | LINE | SUB | FB | Phase distortion (CZ); **FB** feeds the output back into the phase: hard edge to growl. |
| LOFI | CHIP | WAVE | DUTY | CRUSH | - | VIB | BEND | MASK | Chip: PLS TRI SAW NOIS WRAM. **BYTE**: ALGO VAR · MOTN GRIT BEND LOOP. **FLOAT** (floatbeat): ALGO VAR · MOTN FBK BEND LOOP; CHIP sets FULL / 4BIT / 1BIT. |
| VOICE | VOWEL | VOWEL2 | TALK | SHIFT | BUZZ | BREATH | Q | RAND | Formant voice (after klattsch): vowels, talk, breath. |
| WHEEL | REG | SUB | BODY | TOP | PERC | CLICK | DRIVE | ROTOR | Drawbar organ: registrations, percussion, rotary. |
| NOISE | MODE | COLOR | FREQ | PEAK | TRK | DENSE | DRIFT | CRUSH | Coloured and metallic noise, tracked or free. |
| DRUM-X | MORPH | TUNE | TONE | DECAY | NOISE | FM | FOLD | DRIVE | 8 sounds BD SD CP CH OH TM RS CB, sides A / B: **MORPH** morphs the kit. SOUND · SOUND 2 · SOUND 3: LANE SIDE PITCH PMOD · DECAY NOISE COLOR MODE · LANE PMOD DRIVE WAVE. **X-MOD**: RATE DEPTH SHAPE RAND. |

### FILTER · EVERY ENGINE

TYPE | CUT | RES | DRV

**LP · HP · BP · COMB**. COMB rings tuned to the note: CUT moves it ±32 st, RES how much it rings. ANALOG and LOFI: per voice; the others: on the track, following LFO / ENV DEST FLT (DRV = the track's DIST).

### VOICE · PLAYING

VCE | GLD | GLMOD | PRIO

POLY · MONO · LEG · UNI; glide and its mode; LAST / LOW / HIGH priority. **8 voices** shared by the six tracks (MONO and LEGATO keep their note).

### DICE

**SELECT** on EDIT: a random factory preset of the engine, every parameter moved up to a quarter of its range. DRUM-X: both sides of every sound around the factory kit. On HOME (after YES): a new dark piece in a style (DARK MINIMAL .. DEEP BREAKS), every track a role. GLO + G4 undoes.

## MODULATION

### ENV · LFO

ENV

ATK | DEC | SUS | REL

ENV DEST

FLT | PIT | SHP | LOOP

LOOP: held, the envelope cycles from the sustain: an ADSR-shaped LFO.

LFO

RATE | WAVE | PHS | FADE

SIN TRI SAW SQR S&H

LFO DEST

PIT | FLT | SHP | AMP

### MOD MATRIX · 4 SLOTS A TRACK

SLOT | SRC | DST | AMT

SOURCES

LFO · ENV · VEL · KEY · RAND · MODW · AT · EXPR · M1 · M2 · M3 · M4 · STEP

DESTINATIONS

PITCH · CUT · SHP · AMP · PAN · DIST · CHO · DLY · REV · RATE · VIB · E1…E8 · CLIP · PNCH

E1…E8: the engine's own EDIT knobs. CLIP / PNCH: the master.

### MSEQ · MACROS

MSEQ · A CV SEQUENCER PER TRACK

STEP | LVL | LEN | SLEW

16 levels, DIV on MSEQ 2, in time with the transport. Use it from MOD, SRC STEP.

MACROS M1–M4

Hold LFO: the knobs are M1–M4, latched until LFO again. Each session starts with 2 random routes a track; SELECT / F3 roll new ones, G3 clears. By hand: MOD, SRC M1…M4. Saved with the project.

### SEQUENCER

64 steps a track, up to 4 notes a step, piano roll and drum grid. **Hold a step + turn a knob**: a parameter lock. CHANCE (0–100 %) and RATCH (x1–x4) per step. REC + knobs while playing: automation (64 records a section, MOTION page plays / clears them, AUTO LIST edits them).

PATTERN · SHIFT

LEN | DIV | SWG | GATE

OFS | PIT | - | -

### SCALE · ARP

**ROOT and SCALE** are every melodic track's: CHR MAJ MIN DOR MIX PEN MPEN HARM PHRY LYD LOC MEL BLUES WHOLE DIMHW DIMWH. QNT snaps keys or the sequence, TRN transposes, **STRN** (SCL 2) moves every sequence along the scale.

ARP MODES

OFF UP DN UPDN RND ORD REPEAT DNUP UP+8 CONV DIVG PINKY THUMB WALK CHORD **TRNS** (the keys transpose the sequence)

### SONGS

**8 projects × 4 sections (A–D)**, a song of up to 16 rows by bars (SAVE › ARRANGE; END: STOP, x2–x8, LOOP), SLOOP style. Each row can hold a **scene**: mutes, DRUM-X group mutes, macros and MIDI punch-ins. SAVE held plays sections live on the next bar, stores them, records the song (SONG REC). Each section keeps its sounds, steps, motion, kit and its SLICER.

## EFFECTS

### PER TRACK

FX · SENDS

DST | CHO | DLY | REV

DIST

DST | TYPE | TONE | -

SOFT · HARD · FOLD · CRUSH · RECT.

AUTOPAN

DEPTH | RATE | SYNC | WAVE

The track's pan swung L–R around its PAN. SYNC **BEAT**: RATE 4 BARS … 1/32, in time with the bar; **FREE**: in Hz. SIN · TRI · SAW · SQR · S&H, never a click.

### SLICER · ONE ON A BUS

SLCR | PAT | RATE | DEPTH

OFF · GATE · STUT, 16 patterns, 1/8 … 32T. **SL BUS**: T1–T6 and **FX** (the delay / reverb / chorus returns) ON / OFF: everything on = the whole mix chopped after the effects. **SL PITCH**: the repeats' 16-step pitch (±24 st, tape style, QNT SCL to the scale).

### DELAY · CHORUS

DLY

TIME | FDBK | TONE | MIX

DLY 2

WIDE | PITCH | SPRY | -

Synced or free (5 ms – 1.48 s); PITCH / SPRY: a grain delay.

CHORUS

RATE | DEPTH | STER | -

STER: 0 as always, down to mono, up to twice as wide.

### REVERB · FIVE MODELS

TYPE | SIZE | DAMP | PRE

MOD | RATE | FILT | WIDE

| TYPE | WHAT IT IS | SIZE | DAMP | MOD | RATE | WIDE |
|---|---|---|---|---|---|---|
| ROOM | Classic room | decay | darkens | chorus on the tail | its speed | stereo |
| SPRING | Amp spring tank | length · decay | darkens | wobble | wobble speed | stereo |
| SHIMR | Up an octave on every pass; **PRE → AIR** (±24 st: down darker and lower, up airier and higher) | decay | darkens | how much shimmer | — | — |
| RESO | 4 strings tuned to the scale; **PRE → PITCH** (0–14 steps up), **DAMP → FOLW** (OFF / T1–T6: the chord follows that track's notes) | how long they ring | FOLW | past half: the chord opens | — | — |
| CLOUD | Granular wash; SIZE max **freezes** it; **PRE → AIR** (±24 st: the cloud down / up, darker / airier) | decay · freeze | darkens | grains pitched + chorus | grain density | grain spread |

### MASTER

CLIP | PNCH | DUCK | REL

**CLIP** saturates without getting louder. **PNCH** drives the drums: denser body. **DUCK**: the kick pulls the rest down (REL its release). An always-on **leveler** (slow 2:1, –9…+6 dB) before the limiter keeps patches at a steady volume.

PUNCH-IN FX (FX HELD)

REPEAT 1/8 · 1/16 · 1/32, LPF, HPF; on the notes: OCT–/+, ½ TEMPO, DEC–/+, STUTTER, ATK+, ARP, RANDOM. Knobs: FILTER CRUSH THROW DEPTH. Recordable per section.